Shangkun Jiang

Civil and Coastal Engineering, University of Florida

1949 Stadium Rd, Gainesville, FL 32611

+1-352-757-9806| | | | | |

Education

- **University of Florida** *Aug. 2023 - Expected Dec. 2026*
  Ph.D. in Civil Engineering (Transportation); Minor in Statistics Gainesville, FL, USA
  - Advisor: Dr. Xilei Zhao
  - Committee members: Dr. Lily Elefteriadou, Dr. Siva Srinivasan, Dr. Sayar Karmakar
  - Dissertation: Advancing Transportation Systems Analysis with Emerging Mobility Data: From Operational Utility to Data Quality

- **Southeast University** *Sept. 2020 - Jun. 2023*
  M.S.E. in Traffic and Transportation Nanjing, China
  - Advisor: Dr. Peng Liao
  - Dissertation: Research Methods on the Capacity of Inland Waterways Based on Vessel Traffic Flow Theory

- **Dalian University of Technology** *Sept. 2016 - Jun. 2020*
  B.E. in Transportation Engineering Dalian, China
  - Advisor: Dr. Ronghan Yao

Experience

- **University of Florida** *Aug. 2023 - Present*
  Research Assistant, Department of Civil and Coastal Engineering Gainesville, FL, USA
- **Southeast University** *Sept. 2020 - Jun. 2023*
  Research Assistant, School of Transportation Nanjing, China

Research Interests

Transportation: Big data, Data bias, Computational modeling, Travel behavior analysis, Urban traffic

Artificial Intelligence: Causal inference, Machine learning, Deep learning, Statistical analysis

Disaster management: Evacuation, Emergency traffic modeling, Transportation resilience

Research Projects (Selected)

- ***GeoAI-Based Activity Inference from GPS Trajectories*** *May. 2026 - Present*
  University of Florida
  - Developed **GeoAI-Act**, a two-stage GeoAI framework that converts noisy GPS trajectories into semantically labeled stay episodes using Transformer-based stay detection and LLM-based purpose inference.
  - Engineered trajectory, temporal, data-quality, and OSM-based geographic features from large-scale mobility datasets to support robust stay detection and external validation across different regions and data sources.
  - Evaluated the framework against rule-based, clustering-based, and supervised baselines, achieving **0.873 point-level F1** and **0.813 segment-level F1** for stay detection, with improved stay purpose inference using **Llama-3.1 + QLoRA**.

- ***Assessing spatial bias of mobile device location data*** *Sept. 2025 - Present*
  University of Florida
  - Developed a framework to evaluate spatial bias in **large-scale mobility datasets**.
  - Analyzed **10M+ anonymized mobile device** users across Florida.
  - Identified **systematic underrepresentation** in socially disadvantaged and urban areas, highlighting limitations of mobility datasets for transportation analysis.

- ***Analyzing Wildfire Evacuation Behavior Using Facebook Data*** *Sept. 2024 - Feb. 2025*
  University of Florida
  - Analyzed **evacuation behavior** during the 2025 Palisades and Eaton wildfires using **Facebook data**.
  - Quantified **evacuation compliance rates**, travel distances, and destination types during wildfire emergencies.

- Identified **spatial disparities** in evacuation behavior across communities with different socioeconomic variables.
- Demonstrated the use of **large-scale mobility data** to support disaster evacuation transportation analysis.

- ***Impact of Evacuation Orders on Traffic Conditions (Hurricane Ian)*** *May. 2024 - Sept. 2024*
  University of Florida
  - Evaluated **traffic impacts** of evacuation orders during Hurricane Ian using **high-resolution mobility datasets**.
  - Developed methods to estimate **space-mean speed** and **planning time index** from mobile device location data.
  - Applied **causal model** to estimate effects of evacuation orders on traffic performance.
  - Identified a 14.9% decrease in average traffic speed and a 14.6% increase in travel time variability.
- ***Real-Time Urban Traffic Monitoring Using Transit Buses as Probes*** *Jul. 2023 - Dec. 2023*
  University of Florida
  - Developed algorithms to estimate network-level traffic speeds using GTFS realtime transit data.
  - Designed methods for bus segment-trip extraction and speed estimation across large urban transit networks.
  - Validated results against Bluetooth sensor and Google traffic data.
- ***Vessel Traffic Flow Modeling in Inland Waterways*** *Jan. 2023 - June. 2023*
  Southeast University
  - Analyzed vessel traffic flow dynamics using multi-source maritime traffic datasets.
  - Investigated relationships between vessel speed, spacing, and traffic density.
  - Developed a piecewise fundamental diagram model to characterize vessel traffic flow states.

Publications

B=Book Chapter, C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[J] ***Jiang, S.***, Sun, Y., Wong, W.*, Xu, Y., & Zhao, X. (2025). *Real-time urban traffic monitoring using transit buses as probes*. ***Transportation Research Record****, 2679*(2), 219-236. https://doi.org/10.1177/03611981241260708

[J.X] ***Jiang, S.***, Lovreglio, R., Cova, T.J., Park, S., Xu, S., & Zhao, X*. (2026). *Analyzing Wildfire Evacuation Behavior Using Facebook Data: A Case Study of the 2025 Palisades and Eaton Fires*. Accepted for publication in ***Transportation Research Part D: Transport and Environment***. https://doi.org/10.48550/arXiv.2601.01052

[R&R] ***Jiang, S.****, Zhang, X., Wong, W., Park, S., & Zhao, X. (2026). *Examining the short-term causal impact of evacuation orders on traffic conditions: evidence from Hurricane Ian*. Minor revision invited at ***International Journal of Disaster Risk Reduction***.

[B] ***Jiang, S.****, Zhang, X., & Zhao, X. (2026). *Review of multi-source data for disaster management*. In X. Huang, J. Shi, & M. Yang (Eds.), ***Artificial Intelligence in Safety Science and Engineering*** (Chap. 8, pp. 183–213). Elsevier. https://doi.org/10.1016/B978-0-443-36342-9.00008-6

[J] Yang, W., ***Jiang, S.***, Liao, P.*, & Wang, H. (2023). *Evaluation of car-following model for inland vessel-following behavior*. ***Ocean Engineering****, 284*, 115196. https://doi.org/10.1016/j.oceaneng.2023.115196

[J] Liu, L.*, Zhang, X., ***Jiang, S.***, & Zhao, X. (2025). *Hurricane evacuation analysis with large-scale mobile device location data during hurricane Ian*. ***Transportation Research Part D: Transport and Environment****, 139*, 104559. https://doi.org/10.1016/j.trd.2024.104559

[J] Yang, W., Liao, P.*, ***Jiang, S.***, & Wang, H. (2025). *Analysis of vessel traffic flow characteristics in inland restricted waterways using multi-source data*. ***Ocean Engineering****, 317*, 120065. https://doi.org/10.1016/j.oceaneng.2024.120065

[J] Recalde, A.*, Liu, L., Zhang, X., Park, S., ***Jiang, S.***, & Zhao, X. (2026). *Evacuation Destination Choices during Hurricane Ian: A Direct Demand Modeling Approach*. ***Transportation Research Part D: Transport and Environment****, 156*, 105384. https://doi.org/10.1016/j.trd.2026.105384

[C] ***Jiang, S.***, & Jin, C. J.* (2023, February). *Microscopic simulations of traffic congestion in Runyang Bridge: comparisons between two cases*. In Sixth ***International Conference on Traffic Engineering and Transportation System (ICTETS 2022)*** (Vol. 12591, pp. 865-870). SPIE. https://doi.org/10.1117/12.2668553

[S] Garces, S., ***Jiang, S.***, Lovreglio, R.*, Kuligowski, E., Cova, T.J., Xu, S., & Zhao, X. (2025). *A Systematic Literature Review for Wildfire Emergency Management Platforms*. Manuscript submitted for ***Fire Technology***.

[S] Liao, P.*, Yang, W., ***Jiang, S.***, & Wang, H. (2024). *Evaluation of waterway lock service quality in Yangtze Delta: from the perspectives of customer and supplier*. Manuscript submitted for publication in ***Maritime Policy & Management***. https://doi.org/10.48550/arXiv.2410.07132

PRESENTATIONS & TALKS

[1] ***Jiang, S.***, Sun, Y., Wong, W.*, Xu, Y., & Zhao, X. (Jan, 2024). *Real-time Urban Traffic Monitoring By Using Transit Buses As Probes*. ***Lectern*** session on ***Transportation Research Board 103th***, Washington, DC.

[2] ***Jiang, S.****, Lovreglio, R., Cova, T.J., Park, S., Xu, S., & Zhao, X. (Jan, 2026). *Analyzing Wildfire Evacuation Behavior Using Facebook Data: A Case Study of the 2025 Palisades and Eaton fires*. ***Poster*** session on ***Transportation Research Board 105th***, Washington, DC.

[3] ***Jiang, S.****, Zhang, X., & Zhao, X. (Jan, 2026). *A Systematic Review of Multi-Source Data for Disaster Management: Strengths, Limitations, and Integration Strategies*. ***Poster*** session on ***Transportation Research Board 105th***, Washington, DC.

[4] Sun, Y.*, ***Jiang, S.***, Lovreglio, R., Xu, S., Cova, T.J., & Zhao, X. (Jan, 2026). *Using Survey, GPS, and Facebook Data to Analyze Wildfire Evacuation*. ***Poster*** session on ***Transportation Research Board 105th***, Washington, DC.

[5] Park, S., ***Jiang, S.***, Sun, Y., & Zhao, X.* (Jan, 2026). *Developing a Dynamic Social Vulnerability Index for Public Health (DSVI-PH) Using Location-Based Data in Hurricane Beryl*. ***Poster*** session on ***Transportation Research Board 105th***, Washington, DC.

[6] Yang, W.*, ***Jiang, S.***, Liao, P., & Wang, H. (Jan, 2025). *Assessment of Waterway Lock Service Quality: Perspectives from Customers and Suppliers*. ***Poster*** session on ***Transportation Research Board 104th***, Washington, DC.

[7] Bella, G.*, Tree, D., ***Jiang, S.***, Zhao, X., Palm, M., Miller, R., Grajdura, S. (March, 2026). *Understanding Evacuation Dynamics through Diverse Mobility Datasets: Lessons from the Eaton and Palisades Fires*. ***American Association of Geographers (AAG) Annual Meeting 2026***, San Francisco, CA.

[8] Yan, X.*, Recalde, A., ***Jiang, S.***, Zhang, X., Zhao, X. (March, 2026). *Understanding Evacuation Dynamics through Diverse Mobility Datasets: Lessons from the Eaton and Palisades Fires*. ***American Association of Geographers (AAG) Annual Meeting 2026***, San Francisco, CA.

[9] Liu, L.*, Zhang, X., ***Jiang, S.***, & Zhao, X. (Jan, 2025). *Hurricane Evacuation Analysis with Large-scale Mobile Device Location Data*. ***Poster*** session on ***Transportation Research Board 104th***, Washington, DC.

[10] Zheng, W.*, Lee, D., Chen, X., Dong, Z., & ***Jiang, S.***, (Jan, 2024). *Behavior Mechanism of Travel Mode Choice During Large-scale Events*. ***Poster*** session on ***Transportation Research Board 103th***, Washington, DC.

Skills

- **Programming Languages:** Python, R, C, C#
- **Database Systems:** MySQL, MongoDB
- **Transportation Software:** ArcGIS / QGIS, AutoCAD, VISSIM, SUMO, TransCAD
- **Mathematical & Statistical Tools:** STATA, AMOS, SPSS
- **Research Skills:** Academic Writing, Machine Learning, Deep Learning
- **Language:** English (Fluent), Mandarin (Native)

Academic Service

Reviewer for:

- Applied Ocean Research
- Journal of Ocean Engineering and Science
- Transportation Research Board
- Transportation Research Interdisciplinary Perspectives
- Transportation Research Part D: Transport and Environment